

Given: Due:	Further Maths (Core Pure) HW 14	Name: MARK SCHEME
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Chapters 1.4 & 1.5: Complex Numbers (roots of equations)

Q1.

$$f(x) = 9x^3 - 33x^2 - 55x - 25$$

Given that $x = 5$ is a solution of the equation $f(x) = 0$, use an algebraic method to solve $f(x) = 0$ completely.

(5)

Almost all students recognised that $(x - 5)$ was a factor and the majority successfully found the correct quadratic factor, usually by performing algebraic division. Most students were confident at solving the resulting quadratic with complex roots, and when earlier mistakes had been made, students appeared to be aware that solutions appeared as a complex conjugate pair. Very few students ended with only real solutions.

Question Number	Scheme	Marks
	$(x - 5)$ is a factor of $f(x)$ so $f(x) = (x - 5)(9x^2 + \dots)$ $f(x) = (x - 5)(9x^2 + 12x + 5)$ Solve $(9x^2 + 12x + 5) = 0$ to give $x =$ $(x =) -\frac{2}{3} - \frac{1}{3}i, -\frac{2}{3} + \frac{1}{3}i$ or $-\frac{2}{3} \pm \frac{1}{3}i$ or $-\frac{2 \pm i}{3}$ oe (and 5)	M1 A1 M1 A1cao A1ft (5) (5 marks)
	Notes M1: Uses $(x - 5)$ as factor and begins division or process to obtain quadratic with $9x^2$. Award if no working but quadratic factor completely correct. A1: $9x^2 + 12x + 5$ M1: Solves their quadratic by usual rules leading to $x =$ Award if one complex root correct with no working. Award for $(9x^2 + \dots)$ incorrectly factorised to $(3x + p)(3x + q)$, where $ pq = 5$ A1: One correct complex root. Accept any exact equivalent form. Accept single fraction and $\pm$ A1ft: Conjugate of their first complex root.	

Q2.

Given that 4 and $2i - 3$ are roots of the equation

$$x^3 + ax^2 + bx - 52 = 0$$

where a and b are real constants,

(a) write down the third root of the equation,

(1)

(b) find the value of a and the value of b .

(5)

Many candidates were confused by the root $2i - 3$, written with the imaginary part first, and therefore wrote down $2i + 3$ as their third root. They could then score, at best, only 2 marks out of 5. Of those with the correct third root many then scored full marks. Most firstly multiplied out the two brackets containing the complex roots, followed by $(x - 4)$. Others found the correct quadratic factor by using $-(\text{sum})$ and product of roots and just a few used $(x + 3)^2 = -4$. The latter two approaches were less error prone than multiplying out the brackets. It was not uncommon to see pairs of brackets with the wrong signs, due to using $(x + \text{root})$. Other approaches were rarely seen.

Question Number	Scheme	Marks
	EXAM PAPERS PRACTICE	
(a)	$x^3 + ax^2 + bx - 52 = 0$, $a, b \in \mathbb{R}$, 4 and $2i - 3$ are roots $-2i - 3$	$-2i - 3$ seen anywhere in solution for Q6. B1 [1]
(b) Way 1	$(x - (2i - 3))(x - (-2i - 3)) = x^2 + 6x + 13$ or $x = -3 \pm 2i \Rightarrow (x + 3)^2 = -4; = x^2 + 6x + 13 (= 0)$ $(x - 4)(x - (2i - 3)) = x^2 - (1 + 2i)x + 4(2i - 3)$ $(x - 4)(x - (-2i - 3)) = x^2 - (1 - 2i)x + 4(-2i - 3)$ $(x - 4)(x^2 + 6x + 13) \{= x^3 + ax^2 + bx - 52\}$ $a = 2, b = -11$ or $x^3 + 2x^2 - 11x - 52$	Must follow from their part (a). Any incorrect signs for their part (a) in initial statement award M0; accept any equivalent expanded expression for A1. $(x - 3^{\text{rd}} \text{ root})(\text{their quadratic})$. M1 Could be found by comparing coefficients from long division. At least one of $a = 2$ or $b = -11$ A1 Both $a = 2$ and $b = -11$ A1 [5]
(b) Way 2	Sum $= (2i - 3) + (-2i - 3) = -6$ Product $= (2i - 3) \times (-2i - 3) = 13$ So quadratic is $x^2 + 6x + 13$ $(x - 4)(x^2 + 6x + 13) \{= x^3 + ax^2 + bx - 52\}$ $a = 2, b = -11$ or $x^3 + 2x^2 - 11x - 52$	Attempts to apply either $x^2 - (\text{sum roots})x + (\text{product roots}) = 0$ or $x^2 - 2\text{Re}(a)x + \alpha^2 = 0$ $x^2 + 6x + 13$ A1 $(x - 3^{\text{rd}} \text{ root})(\text{their quadratic})$ M1 At least one of $a = 2$ or $b = -11$ A1 Both $a = 2$ and $b = -11$ A1 [5]
(b) Way 3	$(2i - 3)^3 + a(2i - 3)^2 + b(2i - 3) - 52 = 0$ $5a - 3b = 43$ (real parts) and $6a - b = 23$ (imaginary parts) or uses $f(4) = 0$ and $f(a \text{ complex root}) = 0$ to form equations in a and b . So $a = 2, b = -11$ or $x^3 + 2x^2 - 11x - 52$	Substitutes $2i - 3$ into the displayed equation and equates both real and imaginary parts. $5a - 3b = 43$ and $6a - b = 23$ or $16a + 4b = -12$ and $(2i - 3)^3 + a(2i - 3)^2 + b(2i - 3) - 52 = 0$ / $(-2i - 3)^3 + a(-2i - 3)^2 + b(-2i - 3) - 52 = 0$ Solves these equations simultaneously to find at least one of either $a = \dots$ or $b = \dots$ M1 At least one of $a = 2$ or $b = -11$ A1 Both $a = 2$ and $b = -11$ A1 [5]
(b) Way 4	$b = \text{sum of product pairs}$ $= 4(2i - 3) + 4(-2i - 3) + (2i - 3)(-2i - 3)$ $a = -(\text{sum of 3 roots}) = -(4 + 2i - 3 - 2i - 3)$ $a = 2, b = -11$ or $x^3 + 2x^2 - 11x - 52$	Attempts sum of product pairs. All pairs correct o.e. A1 Adds up all 3 roots M1 At least one of $a = 2$ or $b = -11$ A1 Both $a = 2$ and $b = -11$ A1 [5]
(b) Way 5	Uses $f(4) = 0$ $16a + 4b = -12$ $a = -(\text{sum of 3 roots}) = -(4 + 2i - 3 - 2i - 3)$ $a = 2, b = -11$ or $x^3 + 2x^2 - 11x - 52$	Adds up all 3 roots M1 At least one of $a = 2$ or $b = -11$ A1 Both $a = 2$ and $b = -11$ A1 [5]
		6

Q3.

Given that $x = \frac{1}{2}$ is a root of the equation

$$2x^3 - 9x^2 + kx - 13 = 0, \quad k \in \mathbb{R}$$

find

(a) the value of k ,

(3)

(a) the other 2 roots of the equation.

(4)

A variety of methods were used to find k including evaluating $f(0.5)$, long division and inspection. Generally all were quite successful, but those who attempted long division or tried to find the quadratic factor by inspection often made errors. A small number of candidates substituted $x = -\frac{1}{2}$ instead of $x = \frac{1}{2}$ and gained no marks in part (a).

Once $k = 30$ had been successfully achieved, part (b) proved very accessible for most candidates. Once again, the quadratic factor was obtained from a variety of methods, with long division being the most popular but there were often a few sign errors here. Once the quadratic factor had been obtained, candidates went on to find the two complex solutions either by completing the square or by using the formula.

Question Number	Scheme	Notes	Marks
(a)	Ignore part labels and mark part (a) and part (b) together.		
	$f\left(\frac{1}{2}\right) = 2\left(\frac{1}{2}\right)^3 - 9\left(\frac{1}{2}\right)^2 + k\left(\frac{1}{2}\right) - 13$	Attempts $f(0.5)$	M1
	$\left(\frac{1}{4}\right) - \left(\frac{9}{4}\right) + \left(\frac{k}{2}\right) - 13 = 0 \Rightarrow k = \dots\dots$	Sets $f(0.5) = 0$ and leading to $k =$	dM1
	$k = 30$	cao	A1
	Alternative using long division:		
	$2x^3 - 9x^2 + kx - 13 \div (2x - 1)$ $= x^2 - 4x + \frac{1}{2}k - 2$ (Quotient) Remainder $\frac{1}{2}k - 15$	Full method to obtain a remainder as a function of k	M1
	$\frac{1}{2}k - 15 = 0$	Their remainder = 0	dM1
	$k = 30$		A1
	Alternative by inspection:		
	$(2x - 1)(x^2 - 4x + 13) = 2x^3 - 9x^2 + 30x - 13$	First M for $(2x - 1)(x^2 + bx + c)$ or $(x - \frac{1}{2})(2x^2 + bx + c)$ Second M1 for $ax^2 + bx + c$ where $(b = -4 \text{ or } c = 13) \text{ or } (b = -8 \text{ or } c = 26)$	M1dM1
	$k = 30$		A1
			(3)
(b)	$f(x) = (2x - 1)(x^2 - 4x + 13)$ or $\left(x - \frac{1}{2}\right)(2x^2 - 8x + 26)$	M1: $(x^2 + bx \pm 13)$ or $(2x^2 + bx \pm 26)$ Uses inspection or long division or compares coefficients and $(2x - 1)$ or $\left(x - \frac{1}{2}\right)$ to obtain a quadratic factor of this form.	M1
	$x^2 - 4x + 13$ or $2x^2 - 8x + 26$	A1 $(x^2 - 4x + 13)$ or $(2x^2 - 8x + 26)$ seen	A1
	$x = \frac{4 \pm \sqrt{4^2 - 4 \times 13}}{2}$ or equivalent	Use of correct quadratic formula for their 3TQ or completes the square.	M1
	$x = \frac{4 \pm 6i}{2} = 2 \pm 3i$	oe	A1
			(4)
			Total 7

Q4.

$$f(x) = 2x^3 - 6x^2 - 7x - 4$$

(a) Show that $f(4) = 0$

(1)

(b) Use algebra to solve $f(x) = 0$ completely.

(4)

This question was well done by many candidates although there were two particular places in the question where marks were lost. In part (a) virtually all successfully substituted $x = 4$ into $f(x)$ but many failed show sufficient working to justify that $f(4) = 0$. Many incorrectly assumed that $3(4)^3 - 6(4)^2 - 7(4) - 4 = 0$ was enough. In part (b) many candidates successfully established the quadratic factor using either long division, comparing coefficients or inspection and went on to solve the resulting quadratic by using the formula or completing the square. A small number made an attempt at factorising.

There were a surprising number of cases where the final mark was lost when candidates failed to give the real root as well as the complex ones or confused solving with factorising. It was quite common to

see $(x - 4)\left(-\frac{1}{2} + \frac{1}{2}i\right)\left(-\frac{1}{2} - \frac{1}{2}i\right)$ as a conclusion.

Question Number	Scheme	Notes	Marks
(a)	$f(x) = 2x^3 - 6x^2 - 7x - 4$		
	$f(4) = 128 - 96 - 28 - 4 = 0$	$128 - 96 - 28 - 4 = 0$	B1
	Just $2(4)^3 - 6(4)^2 - 7(4) - 4 = 0$ or $2(64) - 6(16) - 7(4) - 4 = 0$ is B0 But $2(64) - 6(16) - 7(4) - 4 = 128 - 96 - 28 - 4 = 0$ or $2(4)^3 - 6(4)^2 - 7(4) - 4 = 4 - 4 = 0$ is B1		
	There must be sufficient working to show that $f(4) = 0$		[1]
(b)	$f(4) = 0 \Rightarrow (x - 4)$ is a factor.		
	$f(x) = (x - 4)(2x^2 + 2x + 1)$	M1: $(2x^2 + kx + 1)$ Uses inspection or long division or compares coefficients and $(x - 4)$ (not $(x + 4)$) to obtain a quadratic factor of this form. A1: $(2x^2 + 2x + 1)$ cao	M1A1
	So, $x = \frac{-2 \pm \sqrt{4 - 4(2)(1)}}{2(2)}$ $(2)\left(x^2 + x + \frac{1}{2}\right) = 0 \Rightarrow (2)\left(x \pm \frac{1}{2}\right) \pm k \pm \frac{1}{2} \neq 0 \Rightarrow x =$	Use of correct quadratic formula for their 3TQ or completes the square.	M1
	Allow an attempt at factorisation provided the usual conditions are satisfied and proceeds as far as $x = ..$		
	$\Rightarrow x = \frac{-2 \pm \sqrt{-4}}{2(2)}$		
	$\Rightarrow x = 4, \frac{-2 \pm 2i}{4}$	All three roots stated somewhere in (b). Complex roots must be at least as given but apply isw if necessary.	A1
			[4]
			5 marks

Q5.

The roots of the equation

$$z^3 - 8z^2 + 22z - 20 = 0$$

are z_1, z_2 and z_3 .

(a) Given that $z_1 = 3 + i$, find z_2 and z_3 .

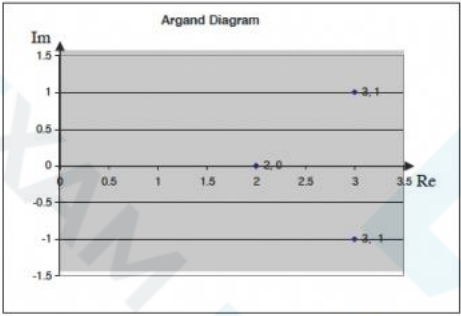
(4)

(b) Show, on a single Argand diagram, the points representing z_1, z_2 and z_3 .

(2)

In part (a) candidates almost always identified the complex conjugate as being another root and through various methods, established the required quadratic factor and went on to find the real root. Surprisingly, some candidates stated that the real root was $z - 2$. There were some instances where candidates simply did not know how to find any other roots and began by substituting $3 + i$ into the given equation and made little progress. Candidates scored less well in part (b) as their Argand diagram gave no indication of scale for either the conjugate pair or the real root.



Question Number	Scheme	Notes	Marks
(a)	$(z_2) = 3 - i$		B1
	$(z - (3 + i))(z - (3 - i)) = z^2 - 6z + 10$	Attempt to expand $(z - (3 + i))(z - (3 - i))$ or any valid method to establish the quadratic factor e.g. $z = 3 \pm i \Rightarrow z - 3 = \pm i \Rightarrow z^2 - 6z + 9 = -1$ $z = 3 \pm \sqrt{-1} = \frac{6 \pm \sqrt{-4}}{2} \Rightarrow b = -6, c = 10$ Sum of roots 6, product of roots 10 $\therefore z^2 - 6z + 10$	M1
	$(z^2 - 6z + 10)(z - 2) = 0$	Attempt at linear factor with their cd in $(z^2 + az + c)(z + d) = \pm 20$ Or $(z^2 - 6z + 10)(z + a) \Rightarrow 10a = -20$ Or attempts $f(2)$	M1
	$(z_3) = 2$		A1
	Showing that $f(2) = 0$ is equivalent to scoring both M's so it is possible to gain all 4 marks quite easily e.g. $z_2 = 3 - i$ B1, shows $f(2) = 0$ M2, $z_3 = 2$ A1. Answers only can score 4/4		(4)
(b)	 First B1 for plotting (3, 1) and (3, -1) correctly with an indication of scale or labelled with coordinates (allow points/lines/crosses/vectors etc.) Allow $i/-i$ for $1/-1$ marked on imaginary axis. Second B1 for plotting (2, 0) correctly relative to the conjugate pair with an indication of scale or labelled with coordinates or just 2		B1 B1 (2)
			Total 6

Q6.

Given that 2 and $5 + 2i$ are roots of the equation

$$x^3 - 12x^2 + cx + d = 0, \quad c, d \in \mathbb{R};$$

(a) write down the other complex root of the equation.

(1)

(b) Find the value of c and the value of d .

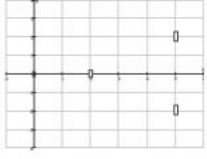
(5)

(c) Show the three roots of this equation on a single Argand diagram.

(2)

This question was generally very well done. Nearly every candidate achieved the complex conjugate in part (a). In part (b) most candidates chose the method of expanding brackets, then equating coefficients to achieve c and d . This was generally very well attempted. Most mistakes were algebraic ones, with methods clearly understood. More confusion reigned amongst those who chose the alternative method. Some substituted 2 and found themselves with one equation in 2 unknowns. Those who substituted a complex root often made mistakes with powers of i or did not use the concept of equating real and imaginary parts. The Argand diagrams in part (c) were good, and most found some way of introducing scale to their diagram, usually by "vectors", co-ordinates, or labelling the axes.



Question Number	EXAM Scheme PRACTICE	Marks
	(a) $5 - 2i$ is a root	B1 (1)
	(b) $(x - (5 + 2i))(x - (5 - 2i)) = x^2 - 10x + 29$ $x^3 - 12x^2 + cx + d = (x^2 - 10x + 29)(x - 2)$ $c = 49, \quad d = -58$	M1 M1 M1 A1, A1 (5)
	(c)  Conjugate pair in 1st and 4th quadrants (symmetrical about real axis) Fully correct, labelled	B1 B1 (2)
	(b) 1 st M: Form brackets using $(x - \alpha)(x - \beta)$ and expand. 2 nd M: Achieve a 3-term quadratic with no i's. (b) <u>Alternative:</u> Substitute a complex root (usually $5 + 2i$) and expand brackets $(5 + 2i)^3 - 12(5 + 2i)^2 + c(5 + 2i) + d = 0$ $(125 + 150i - 60 - 8i) - 12(25 + 20i - 4) + (5c + 2ci) + d = 0$ (2 nd M for achieving an expression with no powers of i) Equate real and imaginary parts $c = 49, \quad d = -58$	M1 M1 M1 A1, A1 [8]